

Ejercicio 1 · Micromanómetro de líquidos inmiscibles — sensibilidad y desplazamiento del menisco

Dos fluidos inmiscibles · Tubo en U · Incremento de presión · Grado de sensibilidad

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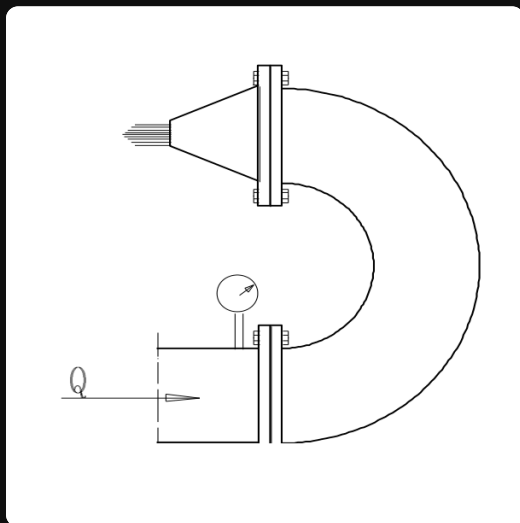
✓ Resuelto

Un micromanómetro de líquidos inmiscibles está formado por dos depósitos abiertos a la atmósfera, unidos mediante un tubo en U. El micromanómetro contiene dos líquidos cuyas densidades relativas son $s_1 = 1$ y $s_2 = 0,9$ respectivamente.

- a) Representar esquemáticamente la situación inicial del micromanómetro cuando los dos depósitos están abiertos a la atmósfera.
- b) Si el menisco de separación se desplaza $j = 5$ cm, calcular el incremento de presión (mm c.a.) que se habrá producido en uno de los depósitos. La sección de los depósitos es muy grande frente a la del tubo ($A \gg a$).
- c) Calcular el grado de sensibilidad del micromanómetro.

Resultados: $\Delta P = 5$ mm c.a.; sensibilidad = 0,98 Pa/mm

FIGURA ORIGINAL (PDF)



 Datos

Variable	Valor	Notas
s_1	$1,0 \rightarrow \rho_1 = 1000 \text{ kg/m}^3$	Fluido 1 (agua)
s_2	$0,9 \rightarrow \rho_2 = 900 \text{ kg/m}^3$	Fluido 2 (aceite ligero)
j	$5 \text{ cm} = 50 \text{ mm}$	Desplazamiento del menisco en el tubo
$A \gg a$	Sección depósito $\gg$ sección tubo	El nivel en los depósitos no varía

Fórmulas

INCREMENTO DE PRESIÓN POR DESPLAZAMIENTO j DEL MENISCO

$$\Delta P = j \cdot (\rho_1 - \rho_2) \cdot g = j \cdot (s_1 - s_2) \cdot \gamma_{\text{agua}}$$

EN MM DE COLUMNA DE AGUA

$$\Delta P [\text{mm c.a.}] = j [\text{mm}] \cdot (s_1 - s_2)$$

GRADO DE SENSIBILIDAD DEL MICROMANÓMETRO

$$\text{Sensibilidad} = \frac{\Delta P}{j} = (s_1 - s_2) \cdot \gamma_{\text{agua}} \quad [\text{Pa/mm}]$$

La ventaja del micromanómetro es que usa fluidos de densidades muy próximas ($s_1 - s_2 \ll 1$): un pequeño ΔP produce un gran desplazamiento j , lo que aumenta la precisión de lectura respecto a un manómetro de mercurio.

Teoría e hipótesis

En un micromanómetro de dos fluidos inmiscibles, la interfaz se forma en el tubo capilar. Cuando se aplica una sobrepresión ΔP en uno de los depósitos, el menisco se desplaza una distancia j en el tubo. Como $A \gg a$, el nivel en los depósitos apenas cambia, y el incremento de presión hidrostática se debe íntegramente al desplazamiento del menisco: $\Delta P = j \cdot (\rho_1 - \rho_2) \cdot g$. Cuanto menor sea la diferencia de densidades, mayor será el desplazamiento para la misma $\Delta P \rightarrow$ mayor sensibilidad.

Resolución

APARTADO B – INCREMENTO DE PRESIÓN PARA $j = 5 \text{ CM}$

¿Por qué? Con el nivel j del líquido manométrico, la diferencia de presión se obtiene directamente de la ecuación manométrica: $\Delta p = (\gamma_2 - \gamma_1) j$. El "incremento" es la variación respecto a la situación de referencia sin flujo.

$$\Delta P = j \cdot (s_1 - s_2) \cdot \gamma_{\text{agua}} = 0,05 \text{ m} \times (1,0 - 0,9) \times 9800 \text{ N/m}^3$$

$$\Delta P = 0,05 \times 0,1 \times 9800 = 49 \text{ Pa}$$

En mm de columna de agua: $\Delta P [\text{mm c.a.}] = j [\text{mm}] \times (s_1 - s_2) = 50 \times 0,1 = \boxed{5 \text{ mm c.a.}}$

APARTADO C – GRADO DE SENSIBILIDAD

¿Por qué? La sensibilidad del manómetro diferencial es $S = \Delta j / \Delta p$. Un valor alto de S indica que pequeñas diferencias de presión producen grandes lecturas j , lo que exige fluidos manométricos con $\gamma_2 \approx \gamma_1$.

$$\text{Sensibilidad} = \frac{\Delta P}{j} = (s_1 - s_2) \cdot \gamma_{\text{agua}} = 0,1 \times 9800 \frac{\text{N}}{\text{m}^3} = 980 \frac{\text{Pa}}{\text{m}} = \boxed{0,98 \frac{\text{Pa}}{\text{mm}}}$$

Equivalente: 1 mm de desplazamiento del menisco corresponde a $\Delta P = 0,98 \text{ Pa} \approx 0,1 \text{ mm c.a.}$

✓ Conclusiones

$\Delta P = 5 \text{ mm c.a.} = 49 \text{ Pa}$ para un desplazamiento del menisco de $j = 5 \text{ cm}$.

Sensibilidad = $0,98 \text{ Pa/mm}$: 10 veces más sensible que un manómetro de agua pura ($\Delta s = 1$).

La baja diferencia de densidades ($s_1 - s_2 = 0,1$) amplifica el desplazamiento del menisco 10 veces respecto a un manómetro convencional de agua.

← Anterior

1 / 9

Siguiente →

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the 1990s, the number of people in the world who are under 15 years of age has increased from 1.1 billion to 1.5 billion, and the number of people aged 65 and over has increased from 0.2 billion to 0.5 billion (United Nations, 1999).

There are a number of reasons why the world's population is ageing. First, the number of people who survive to old age has increased. This is due to a number of factors, including improved medical care, better nutrition, and a decline in the number of people who die from infectious diseases. Second, the number of people who are born has decreased. This is due to a number of factors, including a decline in the number of children born to women, and a decline in the number of people who are born to women who are aged 15 and under.

The ageing of the world's population has a number of implications. First, it will lead to a decline in the number of people who are in the labour force. This will lead to a decline in the number of people who are able to support the elderly. Second, it will lead to a decline in the number of people who are able to support the elderly. This will lead to a decline in the number of people who are able to support the elderly.

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